



UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

Dipartimento di Scienze Fisiche,
Informatiche e Matematiche

13. Assignment 4: Loop Fusion

Linguaggi e Compilatori [I215-011]

Corso di Laurea in INFORMATICA
(D.M.270/04) [16-262]
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Credits

- Barton, IBM Canada, “Revisiting Loop Fusion and its place in the loop transformation framework”
- Absar, ARM L.t.d., “Scalar Evolution Demystified”

Struttura del passo

- Ci serve un `FUNCTION_PASS`, tramite il quale accedere alle informazioni sui loop

```
PreservedAnalyses LoopFusionPass::run(Function &F,  
                                       FunctionAnalysisManager &AM) {  
    LoopInfo &LI = AM.getResult<LoopAnalysis>(F);  
    for (auto *L : LI)  
        ...  
}
```

```
#include "<a href='\"#\"'>llvm/Analysis/LoopInfo.h\"</a>"
```

Dentro il file header del vostro passo

Struttura del passo

- Manipolazione dei Loop
(esempio dal passo LoopVersioning)

```
// Build up a worklist of inner-loops to version. This is necessary as the
// act of versioning a loop creates new loops and can invalidate iterators
// across the loops.
SmallVector<Loop *, 8> Worklist;

for (Loop *TopLevelLoop : *LI)
    for (Loop *L : depth_first(TopLevelLoop))
        // We only handle inner-most loops.
        if (L->isInnermost())
            Worklist.push_back(L);
```

Struttura del passo

- Esplorare i metodi delle classi LoopInfo e LoopInfoBase

Loop Fusion

In order for two loops, L_j and L_k to be fused, they must satisfy the following conditions:

1. L_j and L_k must be adjacent

- *There cannot be any statements that execute between the end of L_j and the beginning of L_k*

2. L_j and L_k must iterate the same number of times

3. L_j and L_k must be control flow equivalent

- *When L_j executes L_k also executes or when L_k executes L_j also executes*

4. There cannot be any negative distance dependencies between L_j and L_k

- *A negative distance dependence occurs between L_j and L_k , L_j before L_k , when at iteration m L_k uses a value that is computed by L_j at a future iteration $m+n$ (where $n > 0$).*

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1) Adiacenza dei loop

Due loop L0 e L1 sono adiacenti se non ci sono basic blocks aggiuntivi nel CFG tra l'uscita di L0 e l'entry di L1.

Se i loop sono **guarded** il successore non loop del guard branch di L0 deve essere l'entry block di L1.

Se i loop **non sono guarded** l'exit block di L0 deve essere il preheader di L1

<https://llvm.org/docs/LoopTerminology.html>

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3) Control flow equivalence

- Due loop L0 e L1 si dicono control flow equivalent se quando uno esegue è garantito che esegua anche l'altro.
- Per determinare la control flow equivalenza ci servono le informazioni di dominanza e postdominanza
 - Se L0 domina L1 e L1 postdomina L0 allora i due loop sono control flow equivalenti

Analisi di dominanza e postdominanza

```
PreservedAnalyses LoopFusePass::run(Function &F,  
FunctionAnalysisManager &AM) {  
    DominatorTree &DT =  
        AM.getResult<DominatorTreeAnalysis>(F);  
    PostDominatorTree &PDT =  
        AM.getResult<PostDominatorTreeAnalysis>(F);  
    ...  
}
```

```
#include "<a href='\"#\"'>llvm/IR/Dominators.h<<  
#include "<a href='\"#\"'>llvm/Analysis/PostDominators.h"
```

Dentro il file header del vostro passo

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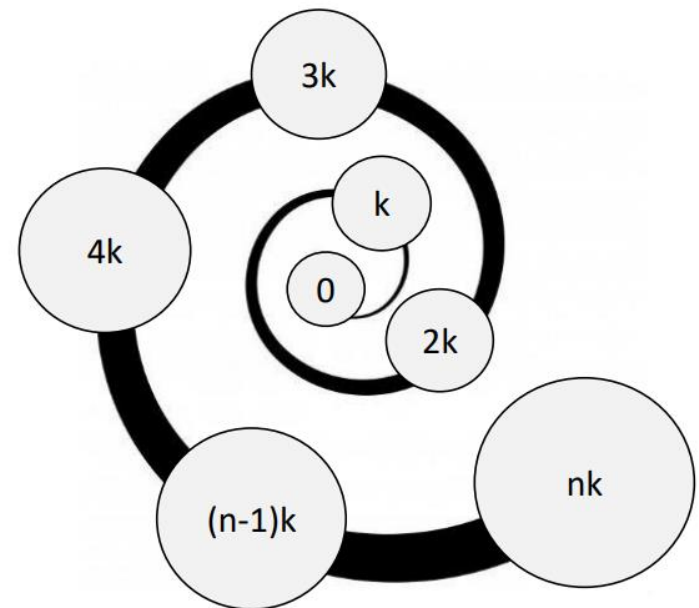
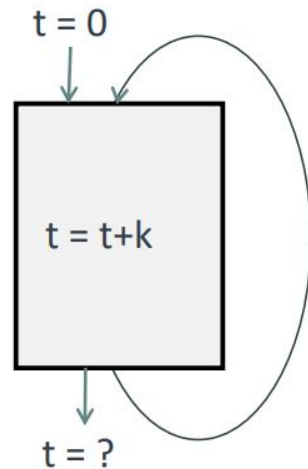
- *A negative distance dependence occurs between L_j and L_k , L_j before L_k , when at iteration m L_k uses a value that is computed by L_j at a future iteration $m+n$ (where $n > 0$).*

Loop Trip Count

- La seconda condizione per la Loop Fusion richiede che si determini il *trip count* dei loop
- Il passo di analisi che, tra le altre cose, determina questo tipo di informazione è la *Scalar Evolution*
- ***Scalar Evolution***: Change in the value of scalar variables over iterations of the loop

Scalar Evolution

```
void foo(int *a, int n, int k) {  
    int t = 0;  
    for (int i = 0; i < n; i++)  
        t = t + k;  
    *a = t;  
}
```



Scalar Evolution

EXAMPLE: Induction variable simplification

```
1. for.body:
2.  %i = phi i32 [ %inc, %for.body ], [ 0, %for.body.preheader ]
3.  %t = phi i32 [ %tk, %for.body ], [ 0, %for.body.preheader ]
4.  %tk = add nsw i32 %t, %k
5.  %inc = add nuw nsw i32 %i, 1
6.  %cmp = icmp slt i32 %inc, %n
7.  br i1 %cmp, label %for.body, label %for.cond.cleanup
...
for.cond.cleanup:
  %tk.final = phi i32 [ 0, %entry ], [ %tk, %for.body]
8.  store i32 %tk.final, i32* %a
...
```

*** IR Dump After Induction Variable Simplification ***

```
...
1'. for.cond.cleanup.loopexit:
2'.  %tk.final = mul i32 %n, %k
3'. store i32 %tk.final, i32* %a
```

Scalar Evolution Analysis

```
PreservedAnalyses LoopFusePass::run(Function &F,  
FunctionAnalysisManager &AM) {  
    ScalarEvolution &SE =  
        AM.getResult<ScalarEvolutionAnalysis>(F);  
    ...  
}
```

```
#include "llvm/Analysis/ScalarEvolution.h"
```

Dentro il file header del vostro passo